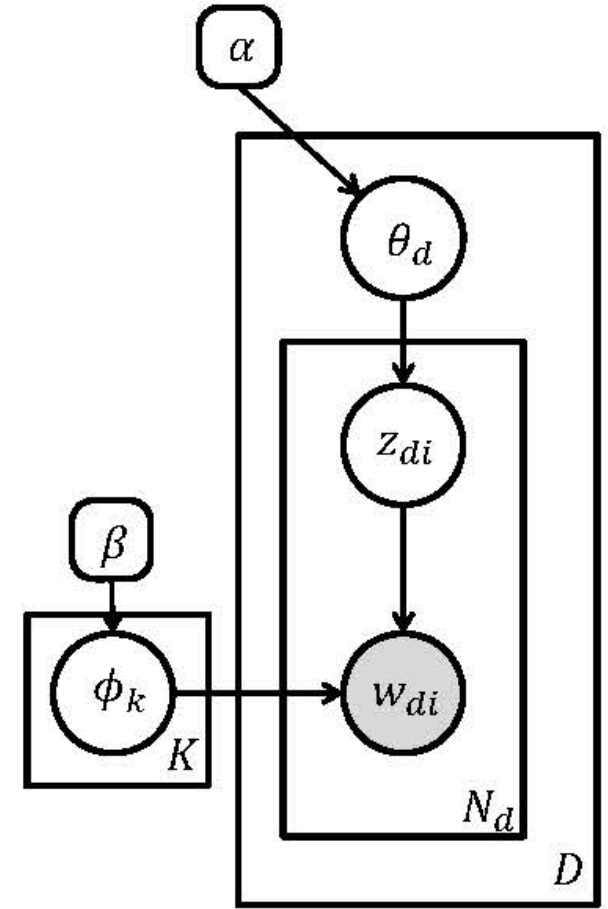

Bayesian Network

HYEONWOO YU

Dept. of Intelligent Robotics, SKKU

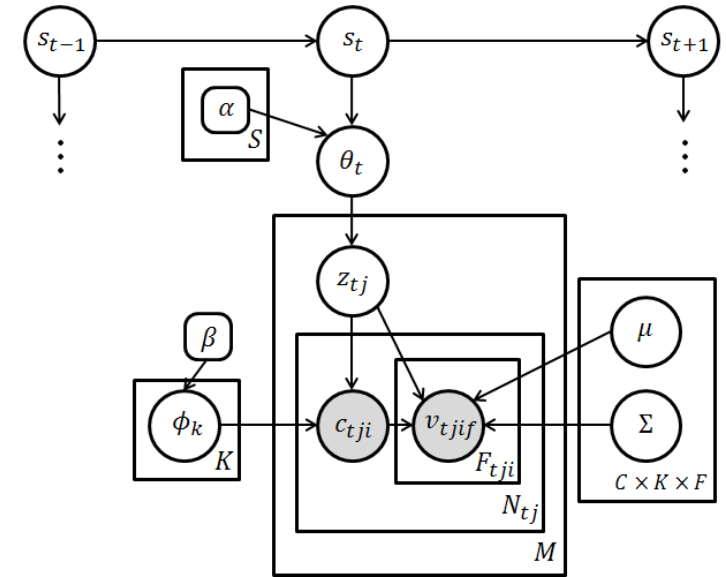


Outline

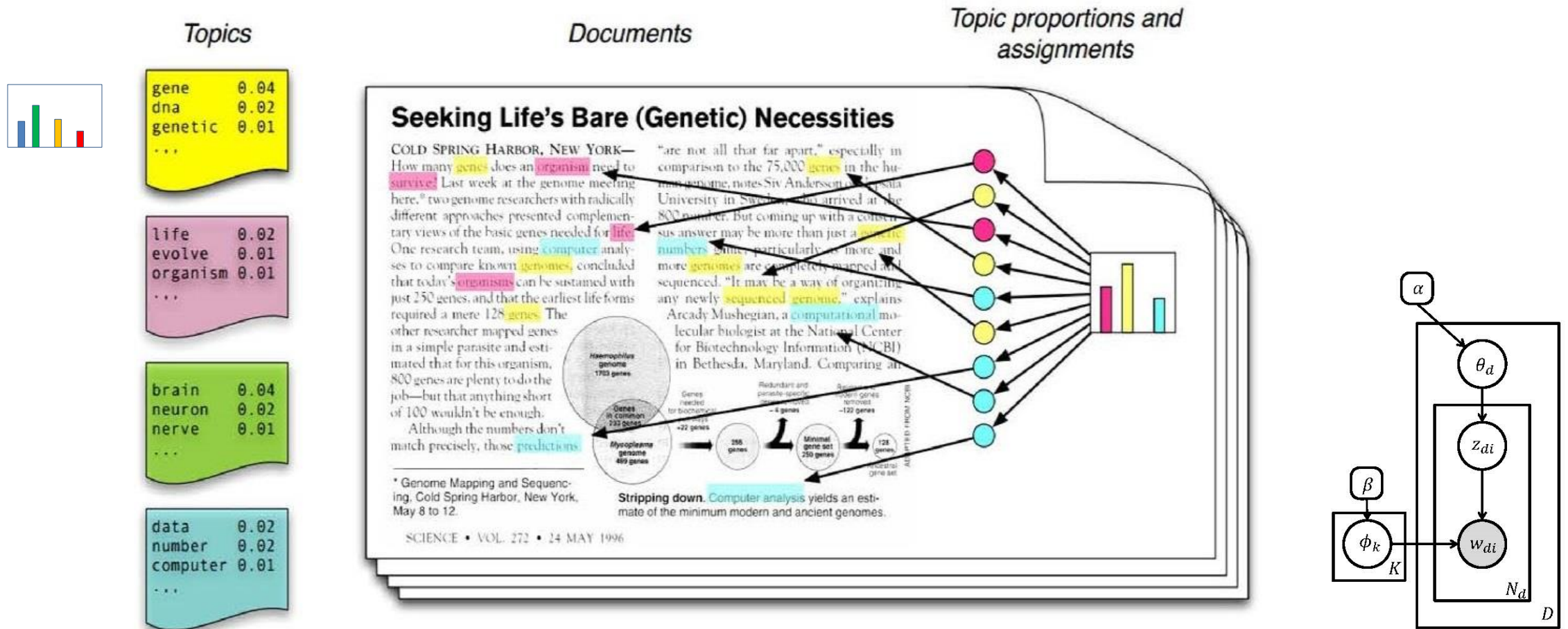
- Application Examples of Bayesian Networks
 - Traffic Pattern Analysis
 - Topic Model (Document Analysis)
- Directed Acyclic Graph
- Conditional Independence
- D-separation
- Bayesian Parameters
- Parameterized Conditional Distributions
- Multinomial, Dirichlet Distribution, Conjugate Prior
- Markov Blanket

Application: Traffic Pattern Analysis

- Surveillance in crowded scenes

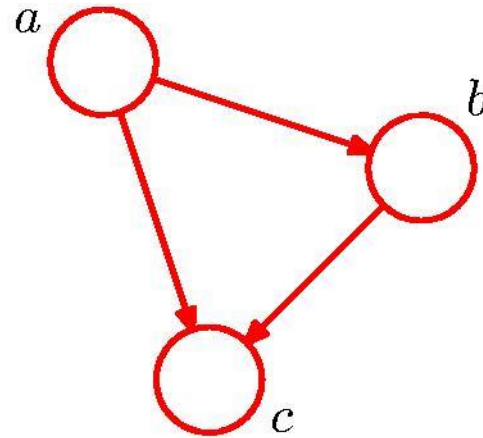


LDA Model (Topic Modelling)



Bayesian Networks

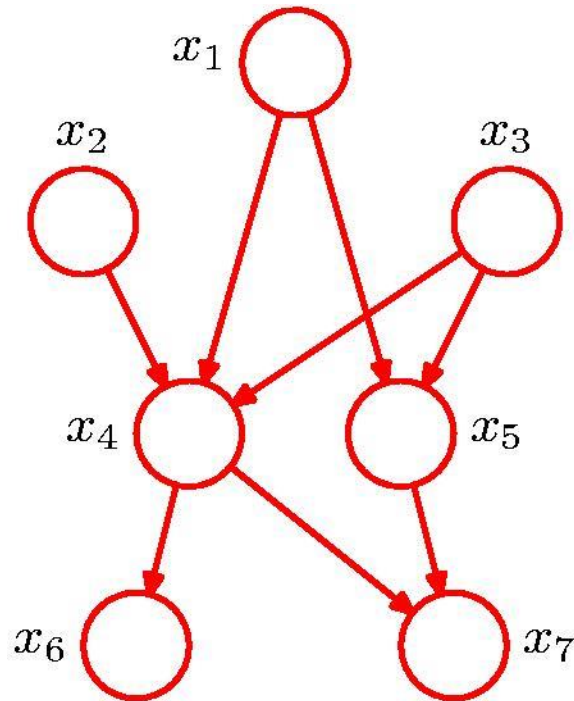
- Directed Acyclic Graph (DAG)



$$p(a, b, c) = p(c|a, b)p(a, b) = p(c|a, b)p(b|a)p(a)$$

$$p(x_1, \dots, x_K) = p(x_K|x_1, \dots, x_{K-1}) \dots p(x_2|x_1)p(x_1)$$

Bayesian Networks



$$p(x_1, \dots, x_7) = p(x_1)p(x_2)p(x_3)p(x_4|x_1, x_2, x_3) \\ p(x_5|x_1, x_3)p(x_6|x_4)p(x_7|x_4, x_5)$$

General Factorization

$$p(\mathbf{x}) = \prod_{k=1}^K p(x_k | \text{pa}_k)$$

Conditional Independence

- a is independent of b given c

$$p(a|b, c) = p(a|c)$$

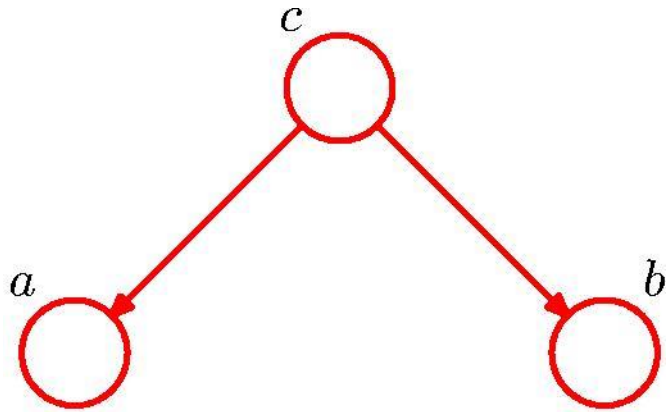
- Equivalently

$$\begin{aligned} p(a, b|c) &= p(a|b, c)p(b|c) \\ &= p(a|c)p(b|c) \end{aligned}$$

- Notation

$$a \perp\!\!\!\perp b \mid c$$

Conditional Independence: Example 1



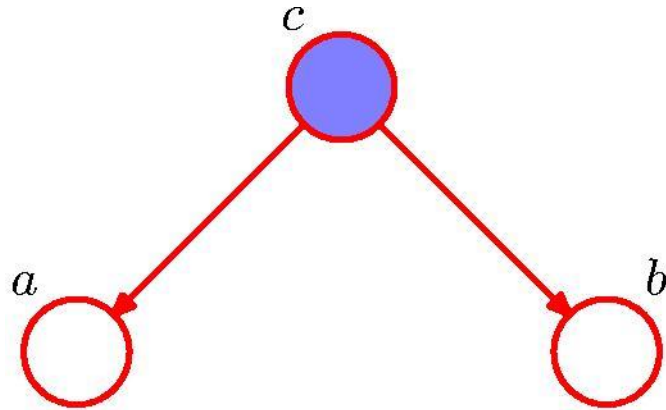
$$p(a, b, c) = p(a|c)p(b|c)p(c)$$

$$p(a, b) = \sum_c p(a|c)p(b|c)p(c)$$

$$a \not\perp b \mid \emptyset$$

U, V , and c are independent. $a = U + c, b = V + c$; a, b independent?

Conditional Independence: Example 1



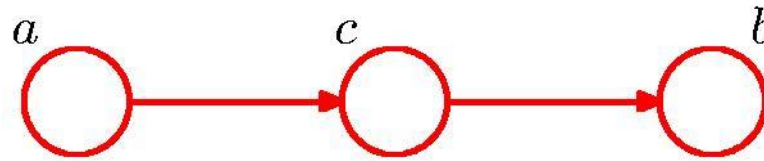
$$p(a, b, c) = p(a|c)p(b|c)p(c)$$

$$\begin{aligned} p(a, b|c) &= \frac{p(a, b, c)}{p(c)} \\ &= p(a|c)p(b|c) \end{aligned}$$

$$a \perp\!\!\!\perp b \mid c$$

U, V , and c are independent. $a = U + c, b = V + c, c = 1$; a, b independent?

Conditional Independence: Example 2



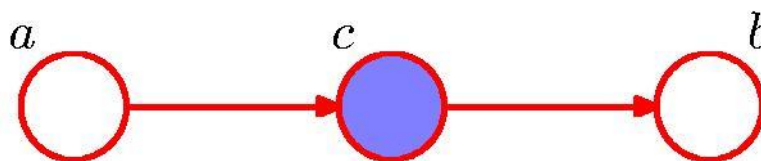
$$p(a, b, c) = p(a)p(c|a)p(b|c)$$

$$p(a, b) = p(a) \sum_c p(c|a)p(b|c) = p(a)p(b|a)$$

$$a \not\perp b \mid \emptyset$$

$$p(b, c|a) = p(c|a)p(b|a, c) = p(c|a)p(b|c)$$

Conditional Independence: Example 2

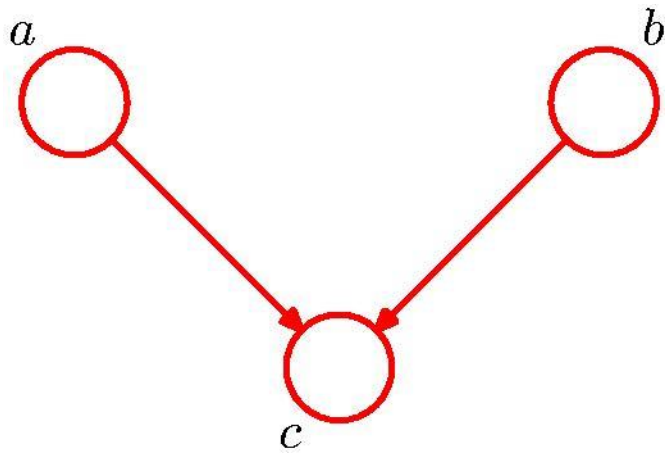


$$\begin{aligned} p(a, b|c) &= \frac{p(a, b, c)}{p(c)} \\ &= \frac{p(a)p(c|a)p(b|c)}{p(c)} \\ &= p(a|c)p(b|c) \end{aligned}$$

$$p(a|c) = \frac{p(c|a)p(a)}{p(c)}$$

$$a \perp\!\!\!\perp b \mid c$$

Conditional Independence: Example 3



$$p(a, b, c) = p(a)p(b)p(c|a, b)$$

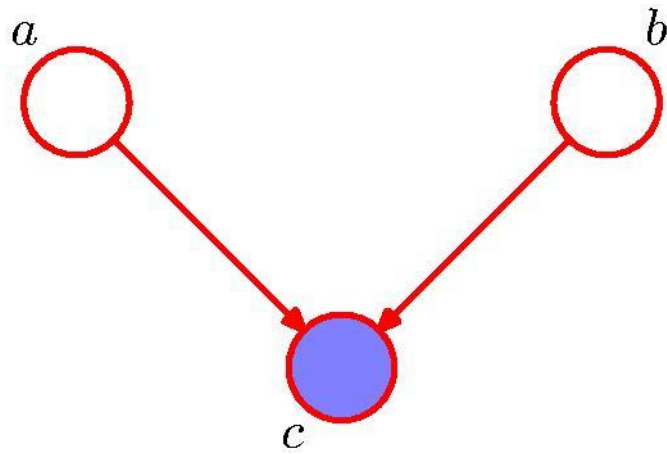
$$p(a, b) = p(a)p(b)$$

$$a \perp\!\!\!\perp b \mid \emptyset$$

Note: this is the opposite of Example 1, with c unobserved.

a and b are independent Bernoulli rvs. $c = a + b$

Conditional Independence: Example 3



$$\begin{aligned} p(a, b|c) &= \frac{p(a, b, c)}{p(c)} \\ &= \frac{p(a)p(b)p(c|a, b)}{p(c)} \end{aligned}$$

$$a \not\perp b \mid c$$

Note: this is the opposite of Example 1, with C observed.

a and b are independent Bernoulli rvs. $c = a + b$

“Am I out of fuel?”

$$\begin{aligned}p(G = 1|B = 1, F = 1) &= 0.8 \\p(G = 1|B = 1, F = 0) &= 0.2 \\p(G = 1|B = 0, F = 1) &= 0.2 \\p(G = 1|B = 0, F = 0) &= 0.1\end{aligned}$$

➤ G is dependent to B and F

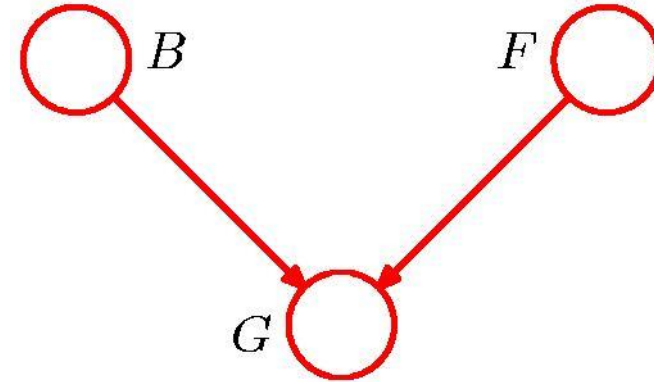
$$p(B = 1) = 0.9$$

$$p(F = 1) = 0.9$$

and hence

$$p(F = 0) = 0.1$$

➤ F is independent to B



B = Battery (0=flat, 1=fully charged)

F = Fuel Tank (0=empty, 1=full)

G = Fuel Gauge Reading
(0=empty, 1=full)

“Am I out of fuel?”

$$p(G = 1 | B = 1, F = 1) = 0.8$$

$$p(G = 1 | B = 1, F = 0) = 0.2$$

$$p(G = 1 | B = 0, F = 1) = 0.2$$

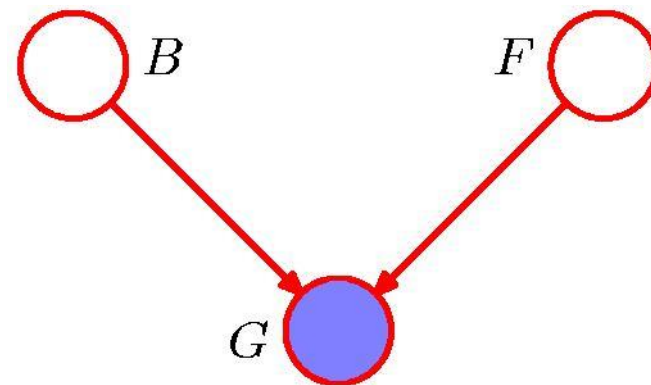
$$p(G = 1 | B = 0, F = 0) = 0.1$$

$$p(B = 1) = 0.9$$

$$p(F = 1) = 0.9$$

and hence

$$p(F = 0) = 0.1$$



Probability of an empty tank increased by observing $G = 0$.

수식화?

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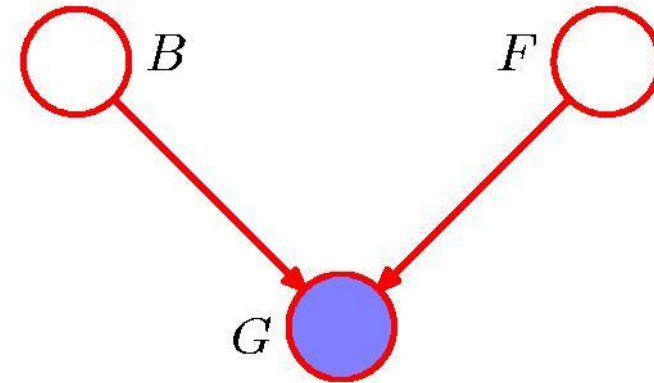
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Probability of an empty tank increased by observing $G = 0$.

$$p(F = 0|G = 0) =$$



“Am I out of fuel?”

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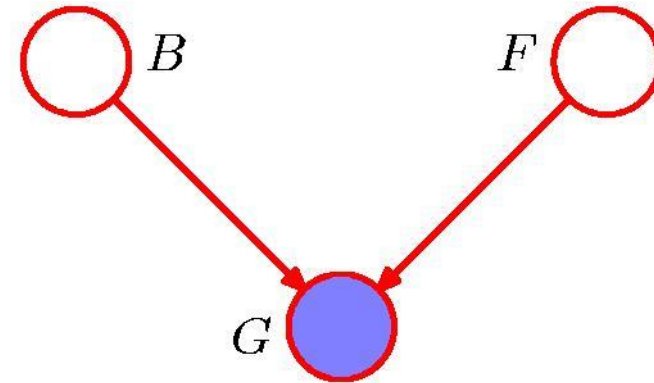
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$$p(F = 1) = 0.9$$

and hence

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Probability of an empty tank increased by observing $G = 0$.

$$p(F = 0|G = 0) = \frac{p(G = 0|F = 0)p(F = 0)}{p(G = 0)}$$

“Am I out of fuel?”

$$p(G = 1|B = 1, F = 1) = 0.8$$

$$p(G = 1|B = 1, F = 0) = 0.2$$

$$p(G = 1|B = 0, F = 1) = 0.2$$

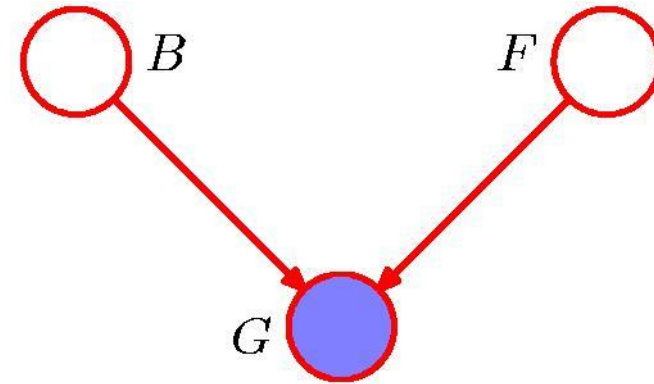
$$p(G = 1|B = 0, F = 0) = 0.1$$

$$p(B = 1) = 0.9$$

$$p(F = 1) = 0.9$$

and hence

$$p(F = 0) = 0.1$$



Probability of an empty tank increased by observing $G = 0$.

$$p(F = 0|G = 0) = \frac{p(G = 0|F = 0)p(F = 0)}{p(G = 0)}$$

$$P(G = 0|F = 0) = 1 - P(G = 1|F = 0) = ?$$

“Am I out of fuel?”

$$p(G = 1|B = 1, F = 1) = 0.8$$

$$p(G = 1|B = 1, F = 0) = 0.2$$

$$p(G = 1|B = 0, F = 1) = 0.2$$

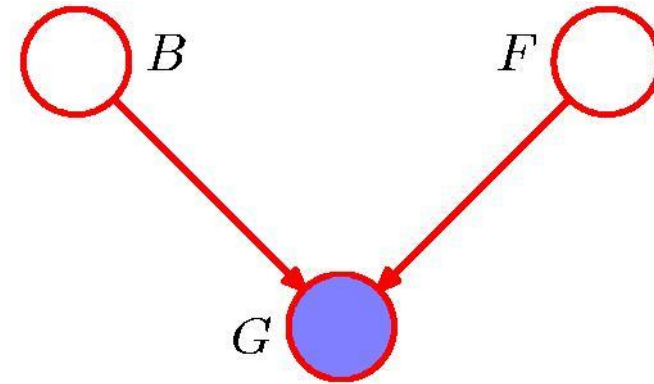
$$p(G = 1|B = 0, F = 0) = 0.1$$

$$p(B = 1) = 0.9$$

$$p(F = 1) = 0.9$$

and hence

$$p(F = 0) = 0.1$$



Probability of an empty tank increased by observing $G = 0$.

$$p(F = 0|G = 0) = \frac{p(G = 0|F = 0)p(F = 0)}{p(G = 0)}$$

$$P(G = 0|F = 0) = 1 - P(G = 1|F = 0) = 1 - \sum_B P(G = 1|B, F = 0)P(B) = 1 - 0.2 \times 0.9 - 0.1 \times 0.1 = 0.81$$

“Am I out of fuel?”

$$p(G = 1|B = 1, F = 1) = 0.8$$

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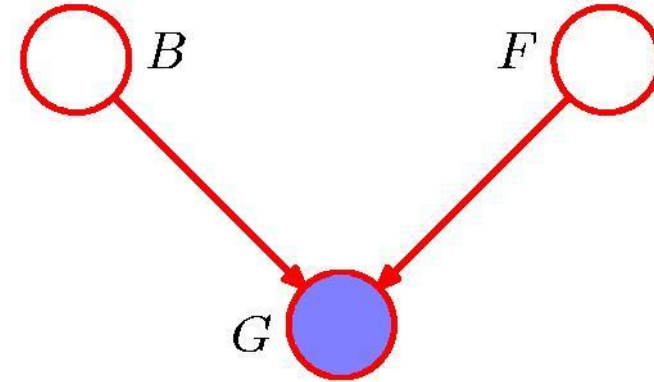
$$p(G = 1|B = 0, F = 0) = 0.1$$

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$$p(F = 1) = 0.9$$

and hence

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Probability of an empty tank increased by observing $G = 0$.

$$p(F = 0|G = 0) = \frac{p(G = 0|F = 0)p(F = 0)}{p(G = 0)}$$

$$P(G = 0|F = 0) = 1 - P(G = 1|F = 0) = 1 - \sum_B P(G = 1|B, F = 0)P(B) = 1 - 0.2 \times 0.9 - 0.1 \times 0.1 = 0.81$$

$$P(G = 0) = 1 - P(G = 1) = 1 - \sum_{B,F} P(G = 1|B, F)P(B, F) = 1 - 0.8 \times 0.81 - 0.2 \times 0.09 - 0.2 \times 0.09 - 0.1 \times 0.01 = 0.315$$

“Am I out of fuel?”

$$p(G = 1|B = 1, F = 1) = 0.8$$

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$$p(G = 1|B = 0, F = 1) = 0.2$$

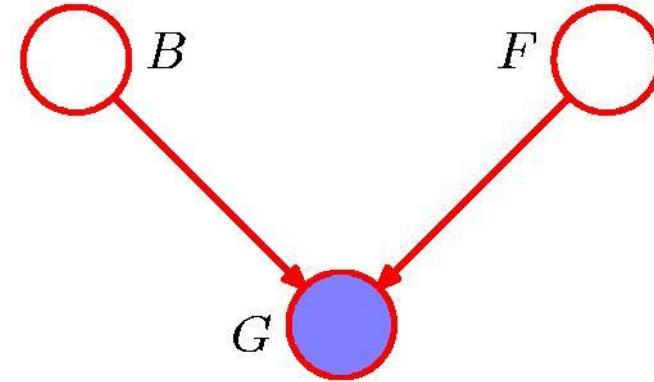
$$p(G = 1|B = 0, F = 0) = 0.1$$

$$p(B = 1) = 0.9$$

$$p(F = 1) = 0.9$$

and hence

$$p(F = 0) = 0.1$$



Probability of an empty tank increased by observing $G = 0$.

$$p(F = 0|G = 0) = \frac{p(G = 0|F = 0)p(F = 0)}{p(G = 0)} = \frac{0.81 \times 0.1}{0.315} = 0.257$$

$$P(G = 0|F = 0) = 1 - P(G = 1|F = 0) = 1 - \sum_B P(G = 1|B, F = 0)P(B) = 1 - 0.2 \times 0.9 - 0.1 \times 0.1 = 0.81$$

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“Am I out of fuel?”

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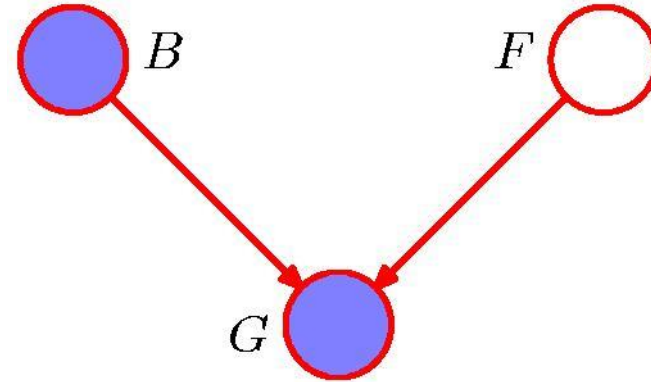
$$p(G = 1|B = 0, F = 0) = 0.1$$

$$p(B = 1) = 0.9$$

$$p(F = 1) = 0.9$$

and hence

$$p(F = 0) = 0.1$$



Probability of an empty tank reduced by observing $B = 0 \& G = 0$
This referred to as “explaining away”.

어떤 영향을 받을 까요? 1) 영향 받지 않는다. 2) Empty 확률이 높아진다. 3) Empty 확률이 낮아진다.

“Am I out of fuel?”

$$p(G = 1|B = 1, F = 1) = 0.8$$

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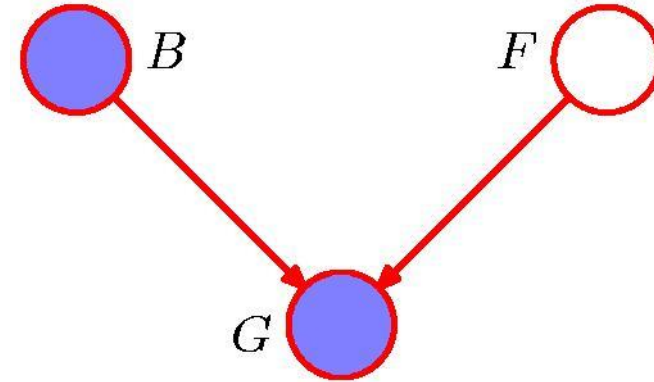
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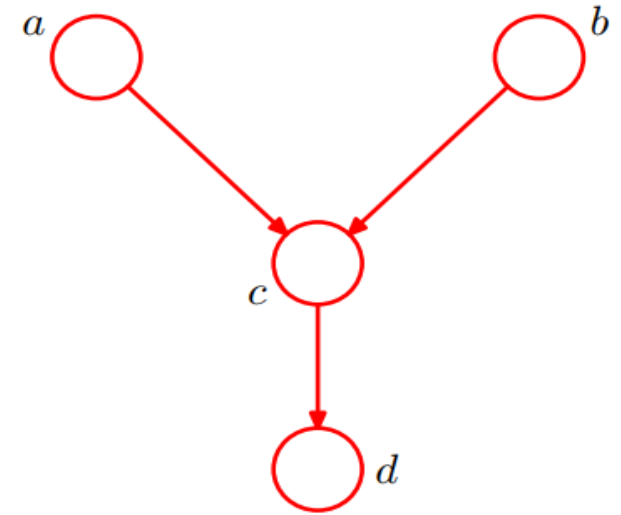
$$\begin{aligned} p(F = 0|G = 0, B = 0) &= \frac{p(G = 0|B = 0, F = 0)p(F = 0)}{\sum_{F \in \{0,1\}} p(G = 0|B = 0, F)p(F)} \\ &\simeq 0.111 \end{aligned}$$

➤ F is dependent to B given G and the empty probability is reduced because the gage becomes less reliable.

Exercise

Answer the following questions for the right-hand Bayesian network.

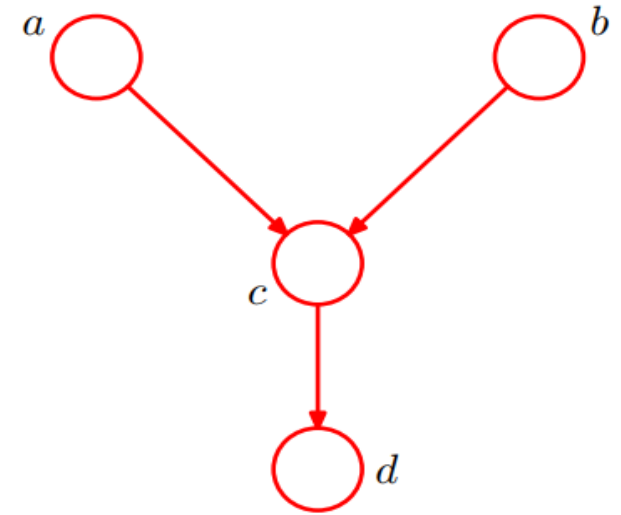
- 1) When any random variables are not observed, show that a and b are independent to each other.
- 2) When d is observed, show that a and b are dependent to each other.



Exercise

Answer the following questions for the right-hand Bayesian network.

- 1) When any random variables are not observed, show that a and b are independent to each other.
- 2) When d is observed, show that a and b are dependent to each other.



1) nothing observed

$$p(a, b) = p(a)p(b) \text{ ?}$$

2) d is observed

$$p(a, b|d) \neq p(a|d)p(b|d) \text{ ?}$$

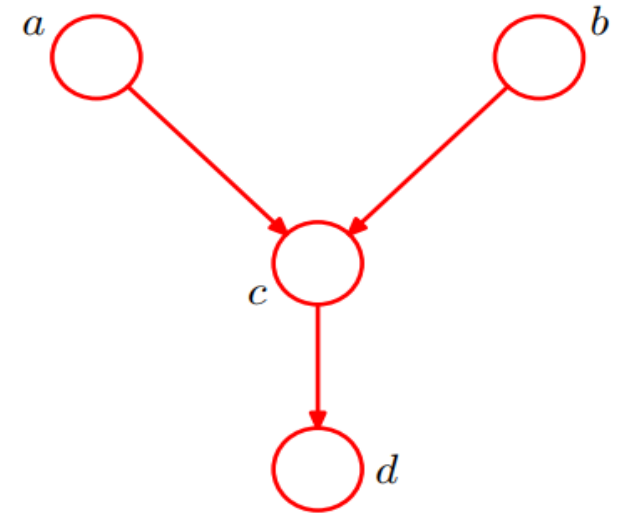
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- 1) When any random variables are not observed, show that a and b are independent to each other.
- 2) When d is observed, show that a and b are dependent to each other.

- 1) Nothing observed

$$p(a, b) = \sum_c \sum_d p(a, b, c, d) =$$



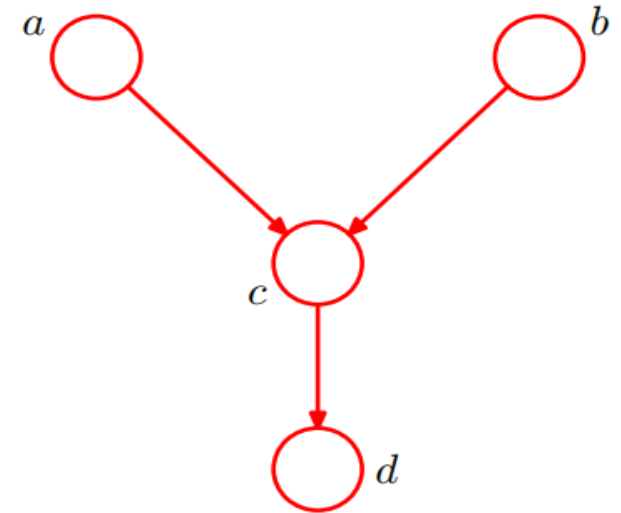
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- 2) When d is observed, show that a and b are dependent to each other.

- 1) Nothing observed

$$p(a, b) = \sum_c \sum_d p(a, b, c, d) = \sum_c \sum_d p(a)p(b)p(c|a, b)p(d|c)$$



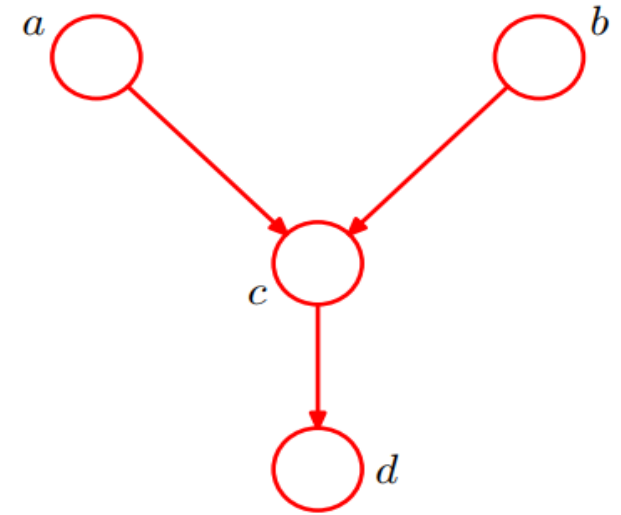
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- 2) When d is observed, show that a and b are dependent to each other.

- 1) Nothing observed

$$\begin{aligned} p(a, b) &= \Sigma_c \Sigma_d p(a, b, c, d) = \Sigma_c \Sigma_d p(a)p(b)p(c|a, b)p(d|c) \\ &= p(a)p(b)\Sigma_c \Sigma_d p(c|a, b)p(d|c) \end{aligned}$$



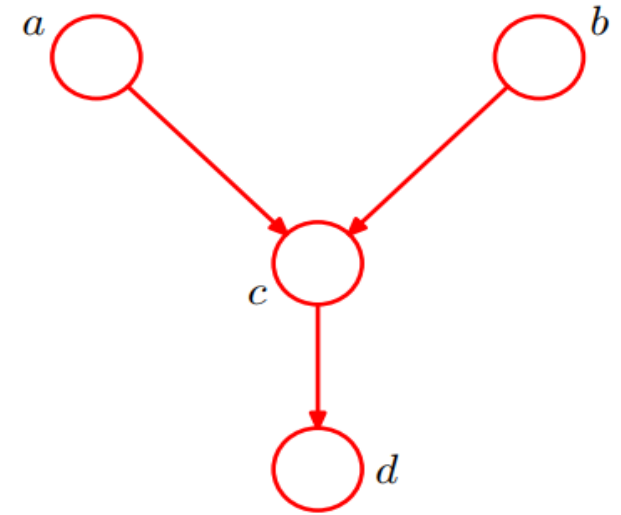
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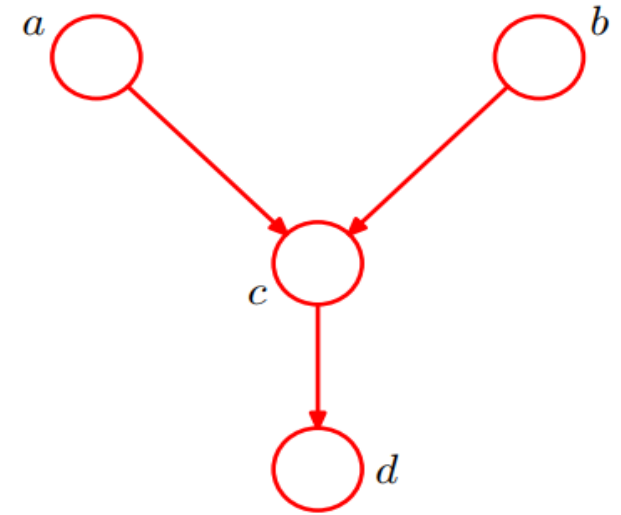
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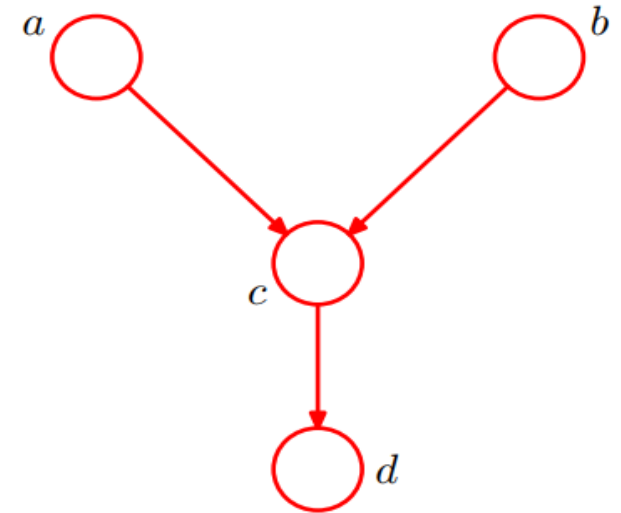
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- 2) When d is observed, show that a and b are dependent to each other.

2) d is observed

$$p(a, b|d) = \frac{p(a, b, d)}{p(d)}$$



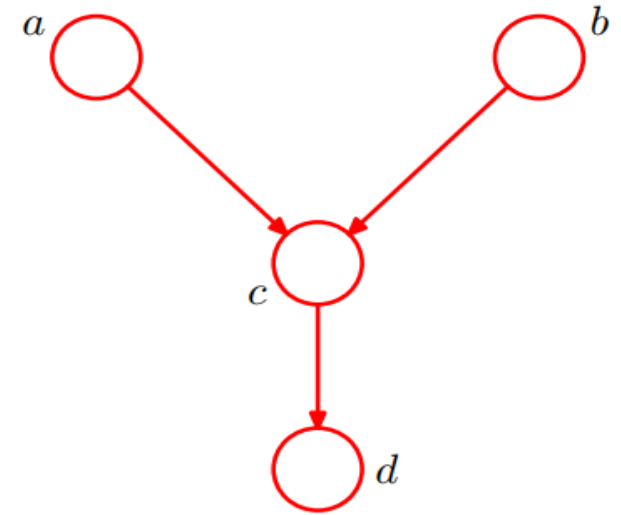
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- 1) When any random variables are not observed, show that a and b are independent to each other.
- 2) When d is observed, show that a and b are dependent to each other.

2) d is observed

$$\begin{aligned} p(a, b|d) &= \frac{p(a, b, d)}{p(d)} \\ &= \sum_c \left\{ \frac{p(a, b, c, d)}{p(d)} \right\} \end{aligned}$$



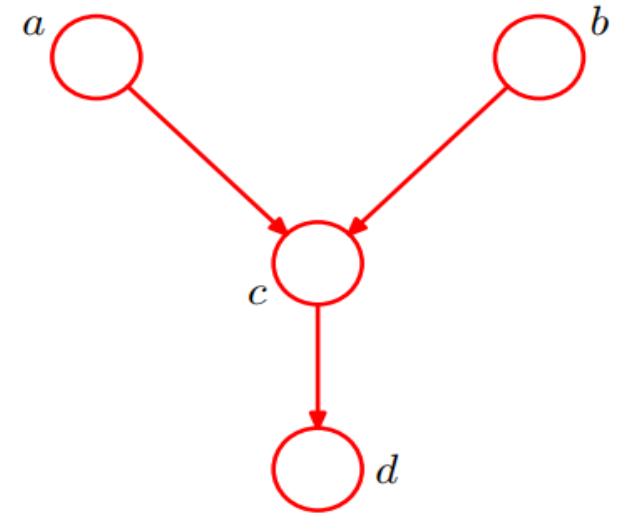
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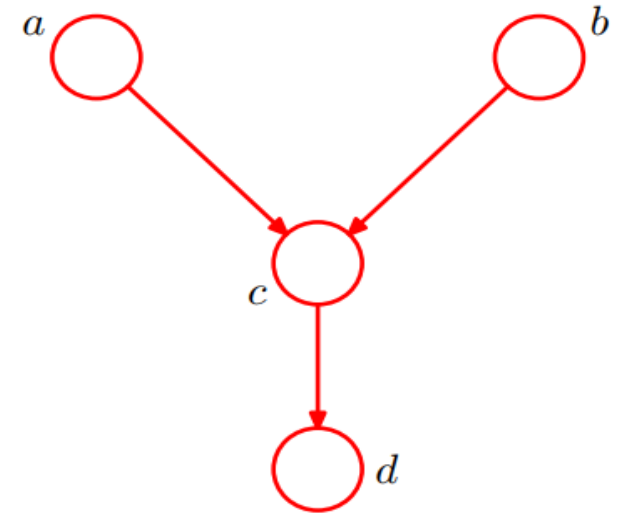
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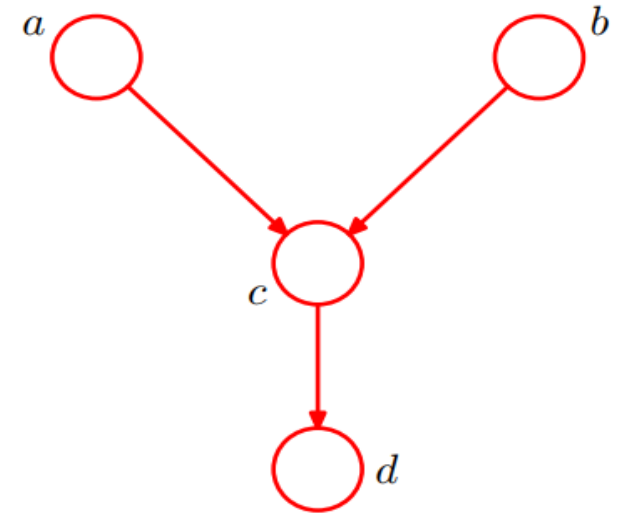
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- 2) When d is observed, show that a and b are dependent to each other.

2) d is observed

$$\begin{aligned} p(a, b|d) &= \frac{p(a, b, d)}{p(d)} \\ &= \sum_c \left\{ \frac{p(a, b, c, d)}{p(d)} \right\} \\ &= \sum_c \left\{ \frac{p(a)p(b)p(c|a, b)p(d|c)}{p(d)} \right\} \\ &= \frac{p(a)p(b)}{p(d)} \sum_c \{ p(c|a, b)p(d|c) \} \\ &= \frac{p(a)p(b)p(d|a, b)}{p(d)} \end{aligned}$$



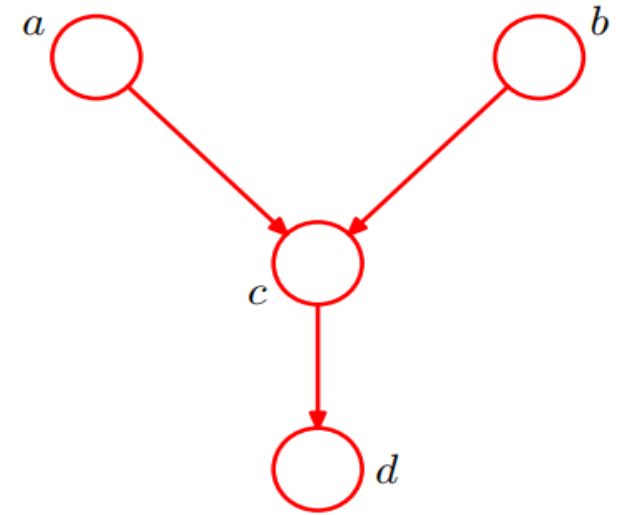
Exercise

Answer the following questions for the right-hand Bayesian network.

- 1) When any random variables are not observed, show that a and b are independent to each other.
- 2) When d is observed, show that a and b are dependent to each other.

2) d is observed

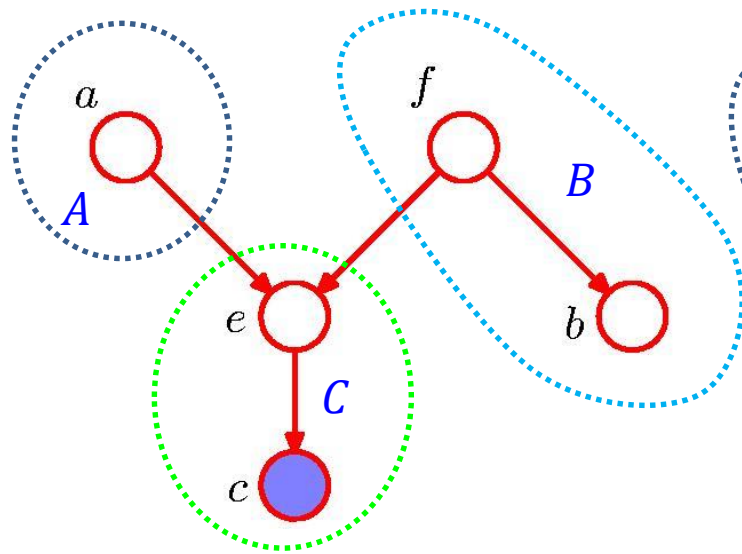
$$\begin{aligned} p(a, b|d) &= \frac{p(a, b, d)}{p(d)} \\ &= \sum_c \left\{ \frac{p(a, b, c, d)}{p(d)} \right\} \\ &= \sum_c \left\{ \frac{p(a)p(b)p(c|a, b)p(d|c)}{p(d)} \right\} \\ &= \frac{p(a)p(b)}{p(d)} \sum_c \{ p(c|a, b)p(d|c) \} \\ &= \frac{p(a)p(b)p(d|a, b)}{p(d)} \neq p(a|d)p(b|d) \end{aligned}$$



D-separation

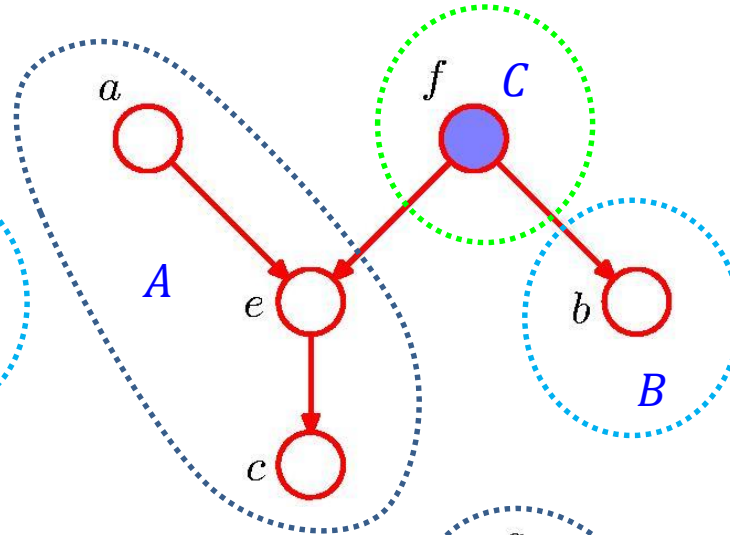
- A , B , and C are non-intersecting subsets of nodes in a directed graph.
- A path from A to B is blocked if it contains a node such that either
 - the arrows on the path meet either head-to-tail or tail-to-tail at the node, and the node is in the set C , or
 - the arrows meet head-to-head at the node, and neither the node, nor any of its descendants, are in the set C .
- If all paths from A to B are blocked, A is said to be d -separated from B by C .
- If A is d -separated from B by C , the joint distribution over all variables in the graph satisfies $A \perp\!\!\!\perp B \mid C$.

D-separation: Example

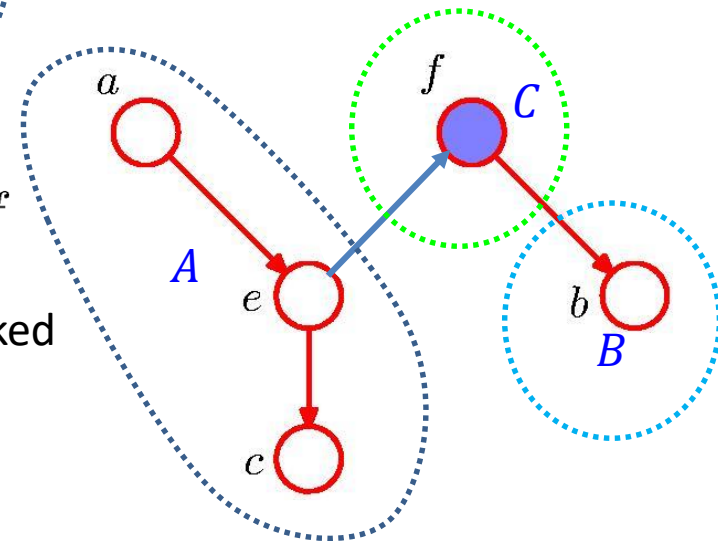


A to B is unblocked

$$a \not\perp\!\!\!\perp b \mid c$$

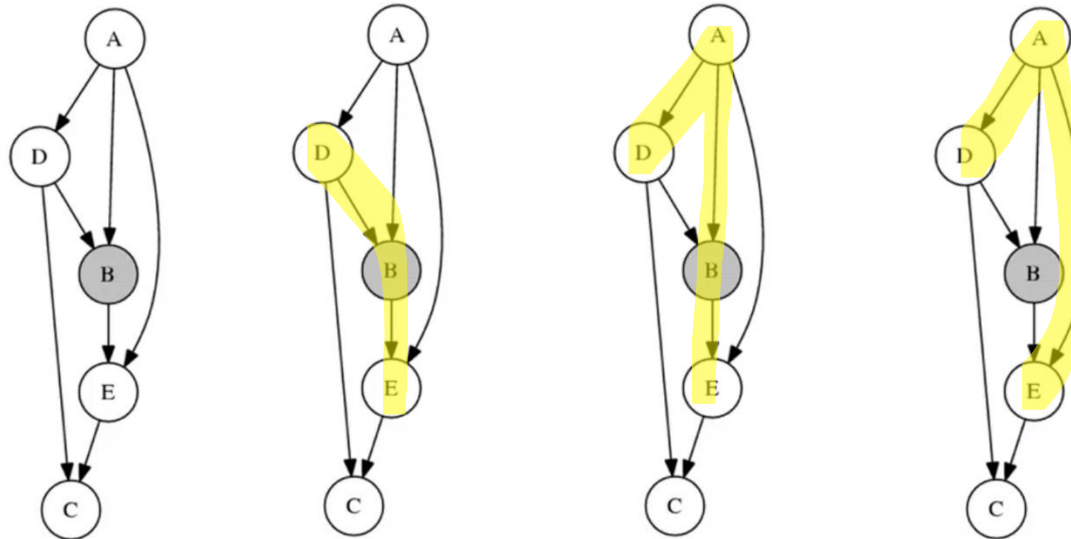


$a \perp\!\!\!\perp b \mid f$
 A to B is blocked
 (d-separated)



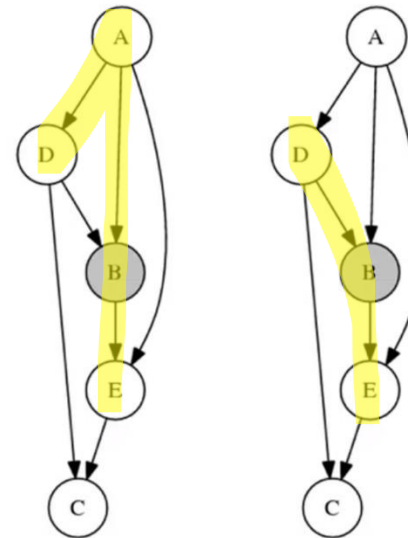
Exercise

When B is observed in the following Bayesian network, decide whether every path from D to E is blocked (d -separated) or not and determine the **dependency between D and E** .



Exercise

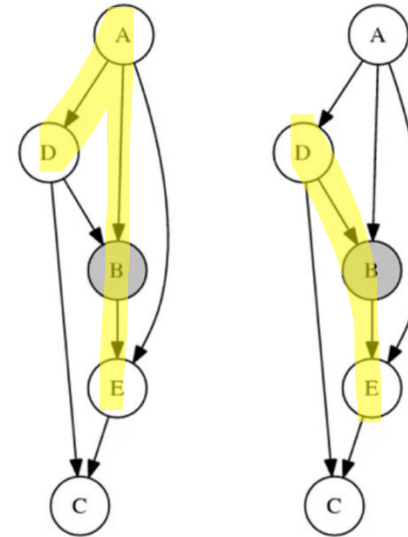
- a. path 1: $(D \leftarrow A \rightarrow B \rightarrow E)$ or $(D \rightarrow B \rightarrow E)$
Is **blocked (d-separated)** or not?



Exercise

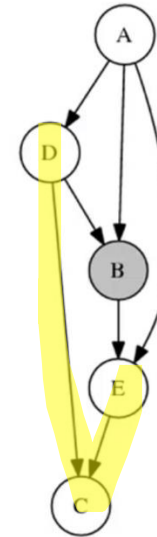
- a. path 1: path via B, i.e., $(D \leftarrow A \rightarrow B \rightarrow E)$ or $(D \rightarrow B \rightarrow E)$
Is **blocked (d-separated)** or not?

Answer: blocked (d-separated),
since the connection in B is **head to tail** and B is observed.



Exercise

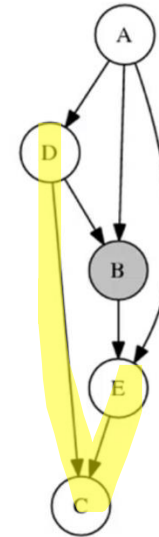
- b. path 2 ($D \rightarrow C \leftarrow E$)
Is **blocked (d-separated)** or not?



Exercise

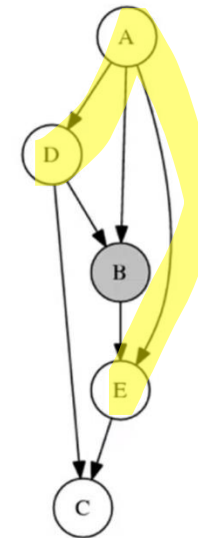
- b. path 2 ($D \rightarrow C \leftarrow E$)
Is **blocked (d-separated)** or not?

Answer: blocked (d-separated),
since the connection in C is **head to head** and C
is not observed.



Exercise

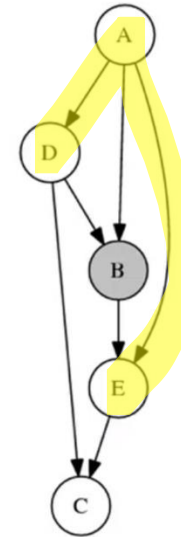
- c. path 3 ($D \leftarrow A \rightarrow E$) :
Is **blocked (d-separated)** or not?



Exercise

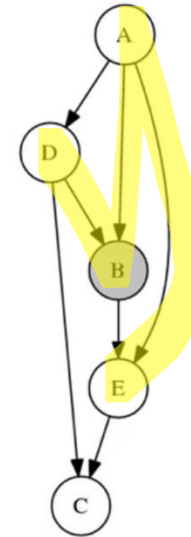
- c. path 3 ($D \leftarrow A \rightarrow E$) :
Is **blocked (d-separated)** or not?

Answer: not blocked,
since the connection in A is **tail to tail** and A is
not observed.



Exercise

- d. path 4 ($D \rightarrow B \leftarrow A \rightarrow E$) :
Is **blocked (d-separated)** or not?

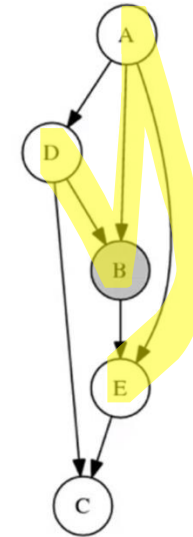


Exercise

- d. path 4 ($D \rightarrow B \leftarrow A \rightarrow E$) :
Is **blocked (d-separated)** or not?

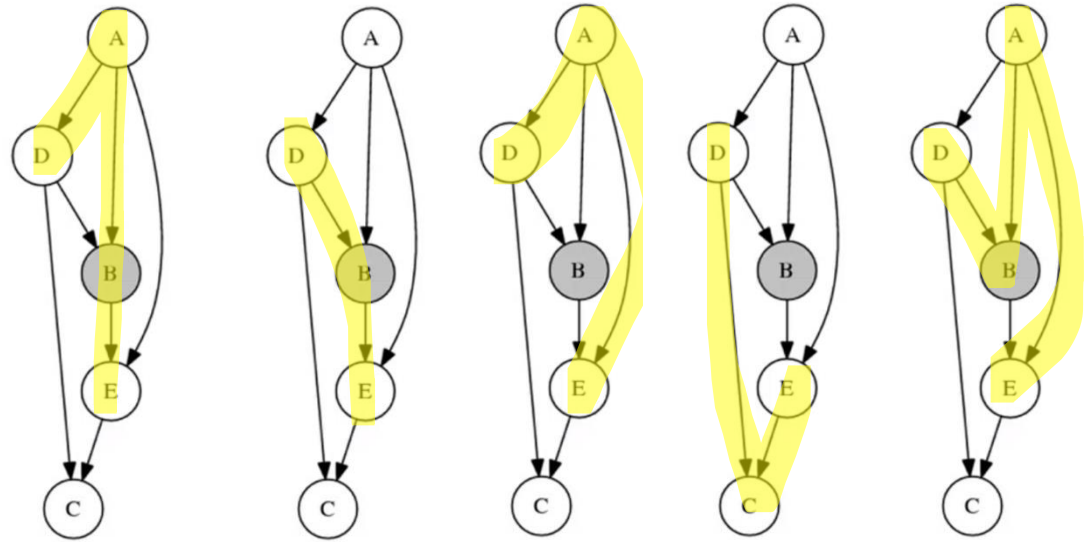
Answer: not blocked,

since the connection in B is **head to head** and B is observed, the path $D \rightarrow B \leftarrow A$ becomes **non-blocking** by B, and
since the connection in A is **tail to tail** and A is not observed, the path $B \leftarrow A \rightarrow E$ becomes **non-blocking**.



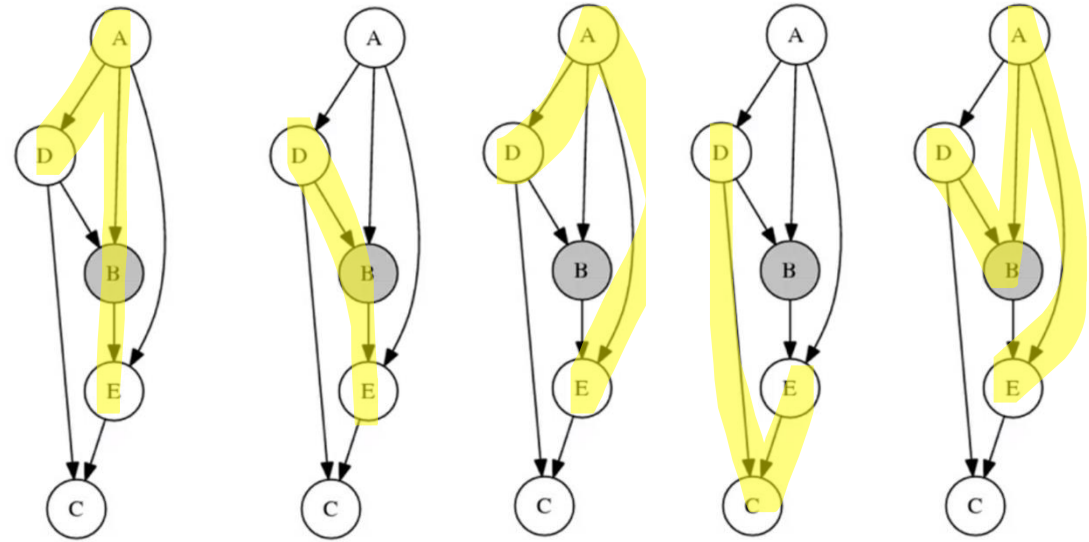
Exercise

- e. There are 4 blocked paths and 1 non-blocked path.
Is **blocked (d-separated)** or not?



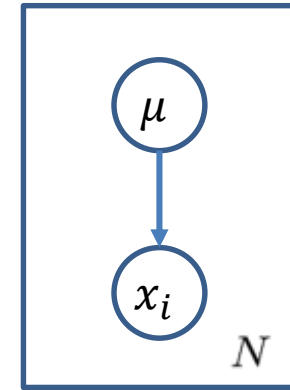
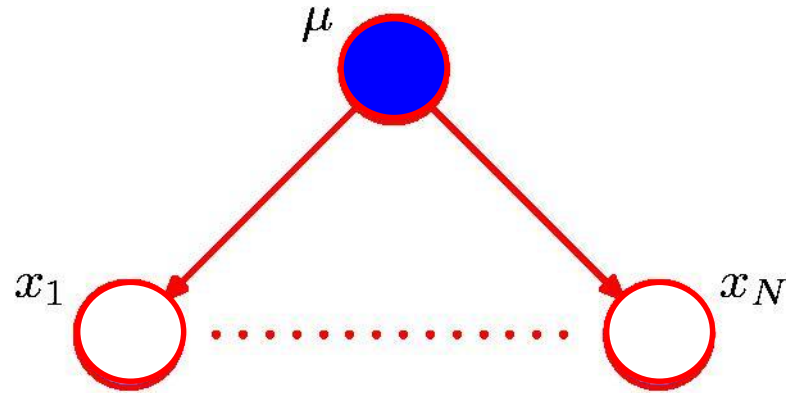
Exercise

- e. There are 4 blocked paths and 1 non-blocked path.
Is **blocked (d-separated)** or not?



Answer: not blocked (d-separated),
since there **exists** at least one non-blocking path.
Thus D and E are **dependent** to each other.

D-separation: I.I.D. Data



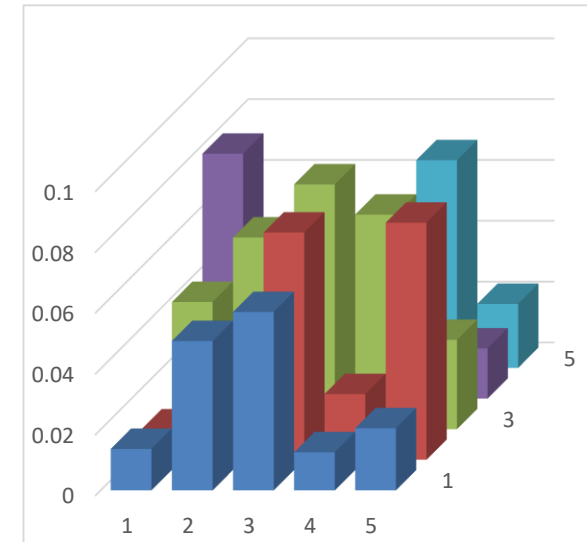
$$p(\mathcal{D}|\mu) = \prod_{n=1}^N p(x_n|\mu)$$

$$p(\mathcal{D}) = \int_{-\infty}^{\infty} p(\mathcal{D}|\mu)p(\mu) d\mu \neq \prod_{n=1}^N p(x_n)$$

Discrete Variables, Multinomial

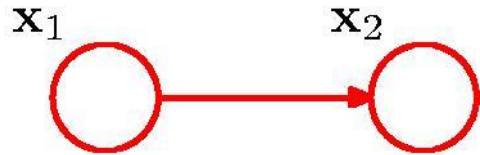
$$p(x_1, \dots, x_K | \mu_1, \dots, \mu_K) = \frac{n!}{x_1! \dots x_K!} \mu_1^{x_1} \dots \mu_K^{x_K}$$

$$p(x_{11}, \dots, x_{1K}, x_{21}, \dots, x_{2K} | \mu_{11}, \dots, \mu_{KK}) = \frac{n!}{x_{11}! \dots x_{1K}!} \frac{n!}{x_{21}! \dots x_{2K}!} \mu_{11}^{x_{11} x_{21}} \dots \mu_{KK}^{x_{1K} x_{2K}}$$



Discrete Variables (1), Multinomial

- General joint distribution: $K^2 - 1$ parameters



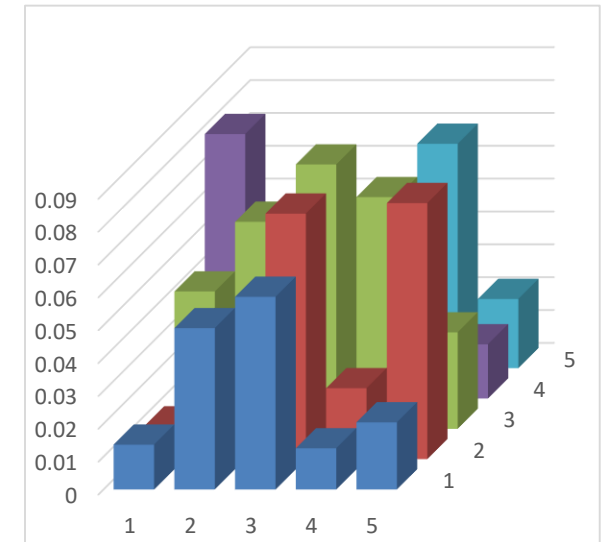
$$p(\mathbf{x}_1, \mathbf{x}_2 | \boldsymbol{\mu}) = \prod_{k=1}^K \prod_{l=1}^K \mu_{kl}^{x_{1k} x_{2l}}$$

- Independent joint distribution: $2(K - 1)$ parameters



$$\hat{p}(\mathbf{x}_1, \mathbf{x}_2 | \boldsymbol{\mu}) = \prod_{k=1}^K \mu_{1k}^{x_{1k}} \prod_{l=1}^K \mu_{2l}^{x_{2l}}$$

- $p(x_1, x_2) = p(x_1 | x_2) p(x_2)$
 $K - 1 + K(K - 1) = K^2 - 1$ parameters
- $p(x_1, x_2) = p(x_1) p(x_2)$
 $K - 1 + K - 1 = 2(K - 1)$ parameters

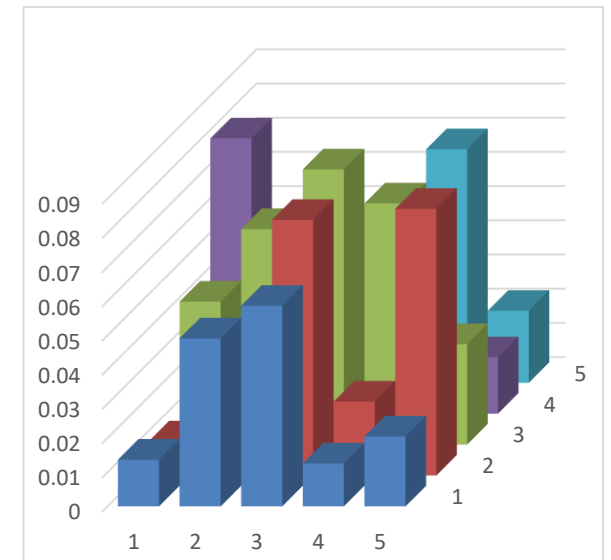


Discrete Variables, Dirichlet

- The posterior distributions are in the same family as the prior probability distribution.

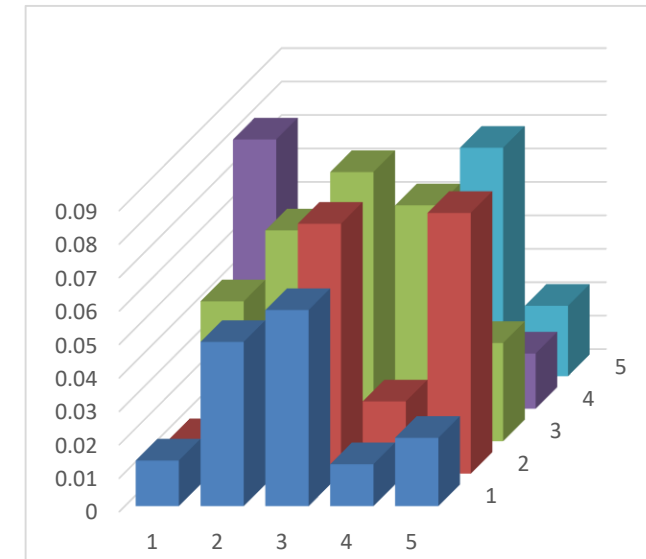
$$p(\mu|x) \propto p(x|\mu)p(\mu)$$

- The prior and posterior are then called **conjugate distributions**, and the prior is called a **conjugate prior** for the likelihood function.
- Dirichlet distribution is a conjugate (prior) distribution to the multinomial distribution.
- Gaussian is a conjugate prior of Gaussian.



Discrete Variables, Dirichlet

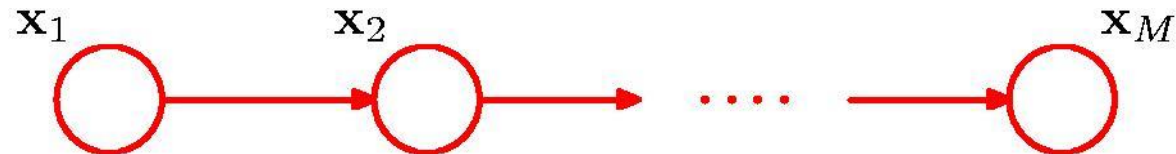
- Posteriori: $p(\mu|x, \alpha) \propto p(x|\mu)p(\mu|\alpha)$
- Mul(K, μ): $p(x_1, \dots, x_K | \mu_1, \dots, \mu_K) = \frac{n!}{x_1! \dots x_K!} \mu_1^{x_1} \dots \mu_K^{x_K}$
- Dir(K, α): $p(\mu_1, \dots, \mu_K | \alpha_1, \dots, \alpha_K) = \frac{\Gamma(\sum_{i=1}^K (\alpha_i - 1))}{\prod_{i=1}^K \Gamma(\alpha_i - 1)} \mu_1^{\alpha_1} \dots \mu_K^{\alpha_K}$
- Parameters: $\alpha_1, \dots, \alpha_K > 0$ (hyper-parameters)
- Support: $\mu_1, \dots, \mu_K \in (0,1)$ where $\sum_{i=1}^K \mu_i = 1$
- Dir($K, c + \alpha$): $p(\mu|x, \alpha) \propto p(x|\mu)p(\mu|\alpha)$
where $c = (c_1, \dots, c_K)$ is number of occurrences
- $E[\mu_k] = \frac{c_k + \alpha_k}{\sum_{i=1}^K (c_i + \alpha_i)}$



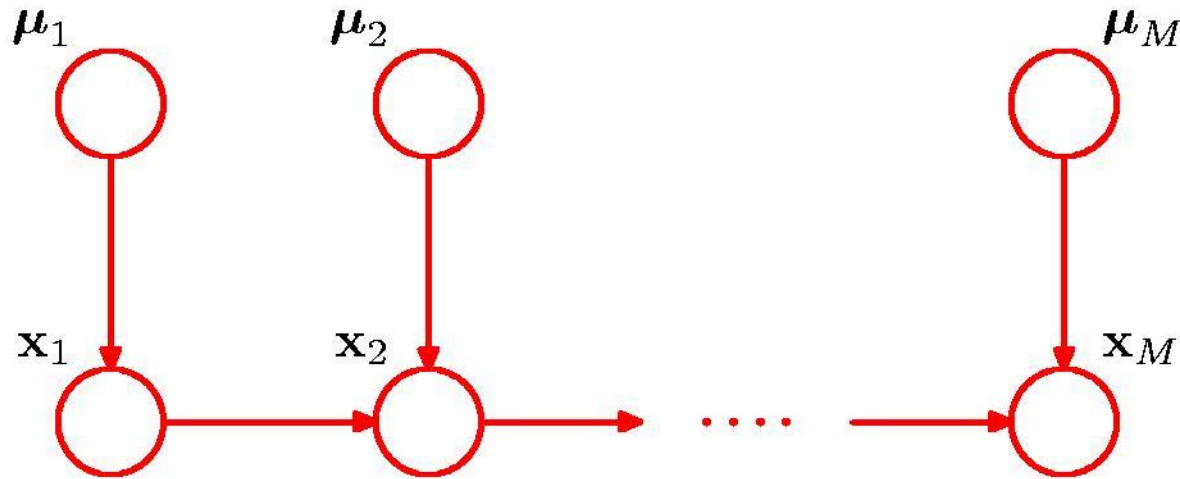
Discrete Variables (2)

- General joint distribution over M variables: $K^M - 1$ parameters
- M -node Markov chain: $K - 1 + (M - 1)K(K - 1)$ parameters

$$p(x_1, \dots) = p(x_1)p(x_2|x_1)p(x_3|x_2)\dots p(x_M|x_{M-1})$$



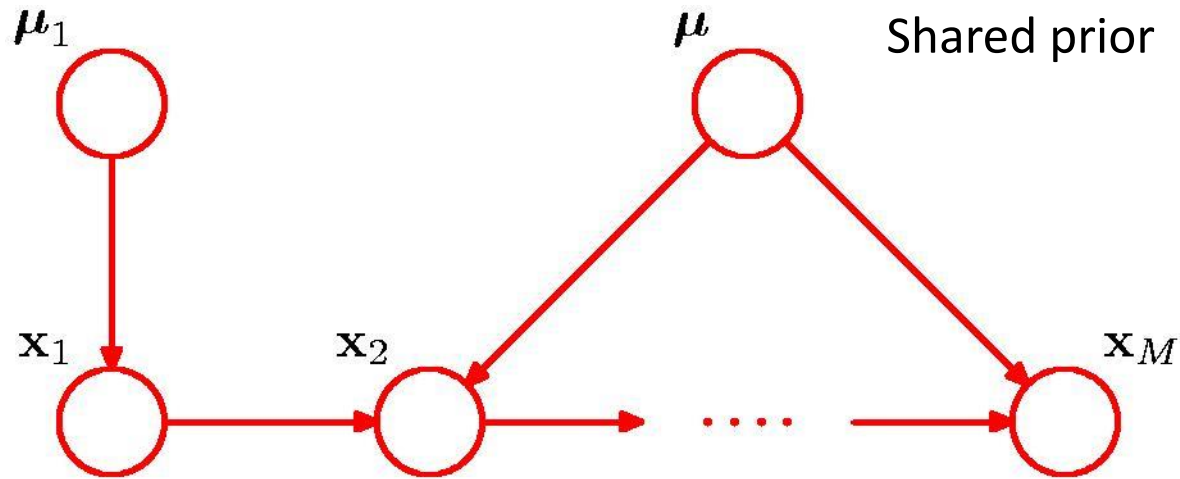
Discrete Variables: Bayesian Parameters (1)



$$p(\{\mathbf{x}_m, \mu_m\}) = p(\mathbf{x}_1 | \mu_1) p(\mu_1) \prod_{m=2}^M p(\mathbf{x}_m | \mathbf{x}_{m-1}, \mu_m) p(\mu_m)$$

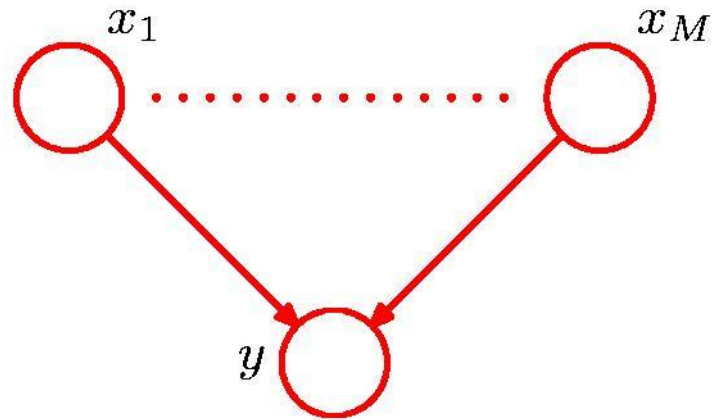
$$p(\mu_m) = \text{Dir}(\mu_m | \alpha_m)$$

Discrete Variables: Bayesian Parameters (2)



$$p(\{\mathbf{x}_m\}, \mu_1, \mu) = p(\mathbf{x}_1 | \mu_1) p(\mu_1) \prod_{m=2}^M p(\mathbf{x}_m | \mathbf{x}_{m-1}, \mu) p(\mu)$$

Parameterized Conditional Distributions



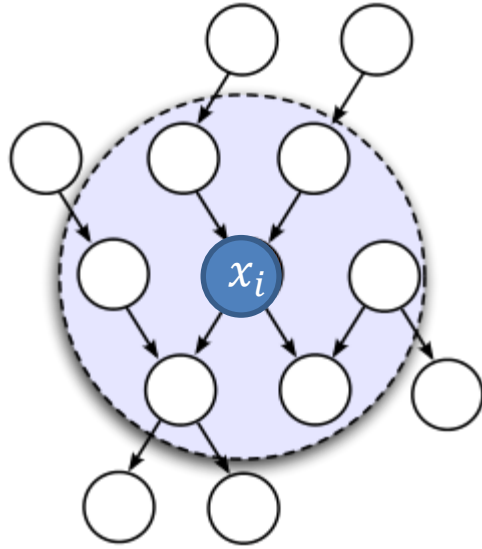
If $x_1, \dots, x_M$ are discrete, K -state variables, $p(y = 1|x_1, \dots, x_M)$ in general has $O(K^M)$ parameters because $p(x_1, \dots, x_M|y = 1)$ requires $K^M - 1$ parameters.

The parameterized form

$$p(y = 1|x_1, \dots, x_M) = \sigma \left(w_0 + \sum_{i=1}^M w_i x_i \right) = \sigma(\mathbf{w}^T \mathbf{x})$$

requires only $M + 1$ parameters (actually this can not model a probability distribution).

The Markov Blanket



$$\begin{aligned} p(\mathbf{x}_i | \mathbf{x}_{\{j \neq i\}}) &= \frac{p(\mathbf{x}_1, \dots, \mathbf{x}_M)}{\int p(\mathbf{x}_1, \dots, \mathbf{x}_M) d\mathbf{x}_i} \\ &= \frac{\prod_k p(\mathbf{x}_k | \text{pa}_k)}{\int \prod_k p(\mathbf{x}_k | \text{pa}_k) d\mathbf{x}_i} \\ &= \prod_{k \in MB} p(x_k | \text{pa}_k) \end{aligned}$$

Any factor $p(x_k | \text{pa}_k)$ that does not have any functional dependence on x_i can be taken outside the integral over x_i , and will therefore cancel between numerator and denominator.

Summary

- Bayesian Networks
- Directed Acyclic Graph
- Conditional Independence
- D-separation
- Bayesian Parameters
- Parameterized Conditional Distributions
- Multinomial, Dirichlet Distribution, Conjugate Prior
- Markov Blanket